ME 165
Basic Mechanical Engineering

Lecture 09

Kinetics of Particles

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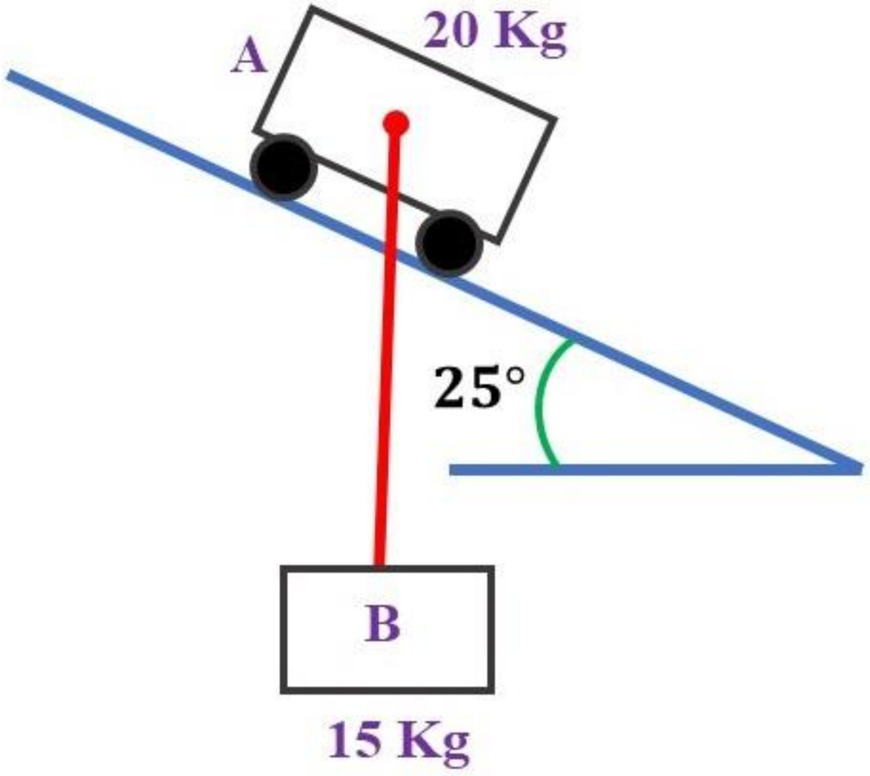
Equation of Motion

Problem:

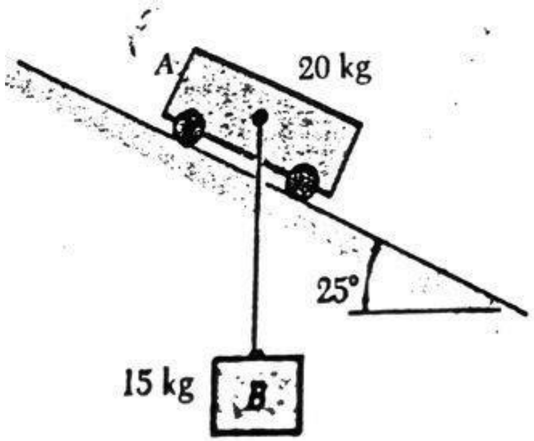
A 15-kg block ***B*** is suspended from a 2.5 m cord attached to a 20-kg cart ***A.*** Neglecting friction, determine

(a) the acceleration of the cart,

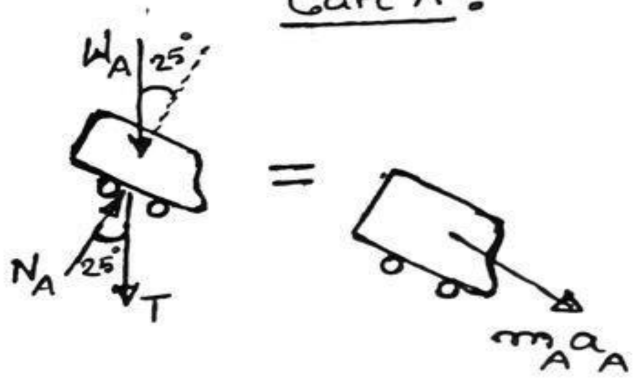
(b) the tension in the cord,
immediately after the system is released from rest in the position shown.



Equation of Motion



Cart 'A':



$$+\searrow \Sigma F = m_A a_A$$

$$\Rightarrow (T + W_A)\sin 25^\circ = 20\, a_A \quad \text{—(1)}$$

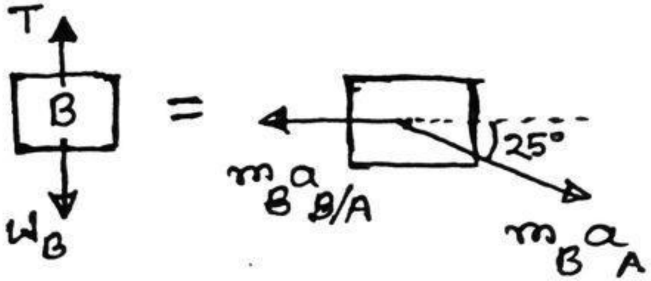
Given,

$W_A = (20 \times 9.81)\ N$

$W_B = (15 \times 9.81)\ N$

$(a)\ a_A = ?\ (b)\ T = ?$

Block 'B':



$$\overset{+}{\rightarrow} \Sigma F = m_B a_B$$

$$\Rightarrow 0 = -m_B a_{B/A} + m_B a_A \cos 25^\circ \text{ —}$$

$$\Rightarrow a_{B/A} = a_A \cos 25^\circ \quad \text{—(ii)}$$

$$+\downarrow \Sigma F = m_B a_B$$

$$\Rightarrow W_B - T = m_B a_A \sin 25^\circ$$

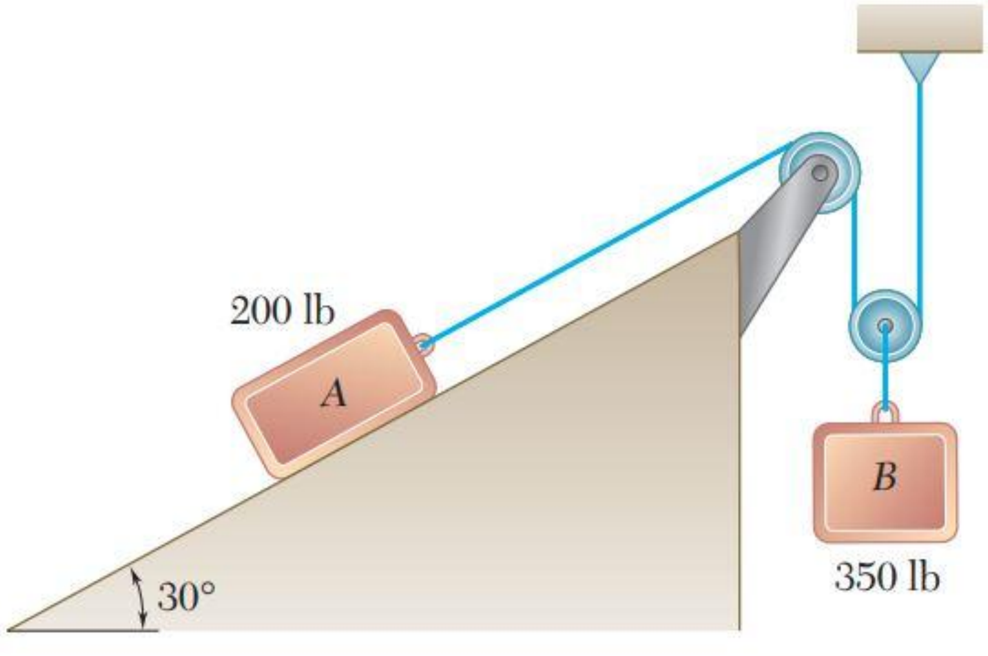
$$\Rightarrow (W_B - T)\sin 25^\circ = 15\, a_A (\sin 25^\circ)^2 \quad \text{—(iii)}$$

Equation of Motion

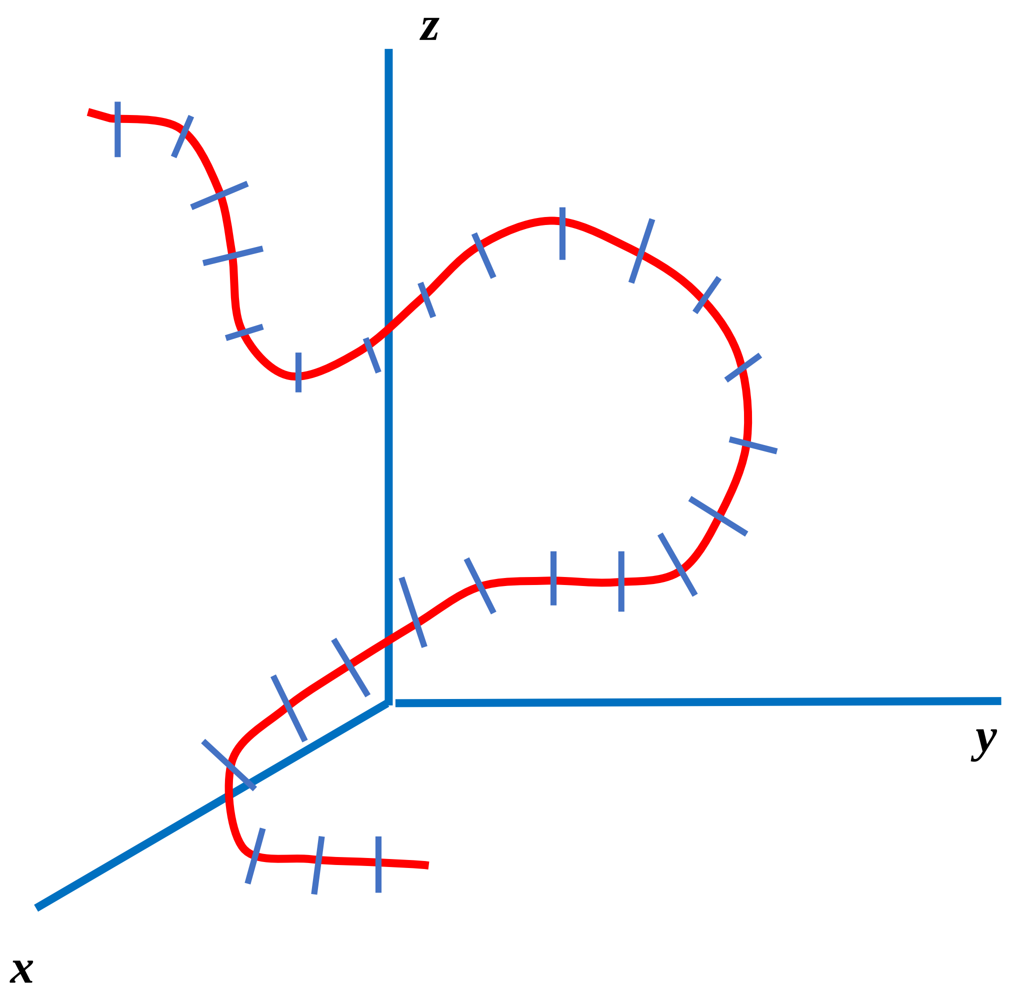
Practice Problem:

The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and between block **A** and the incline, determine

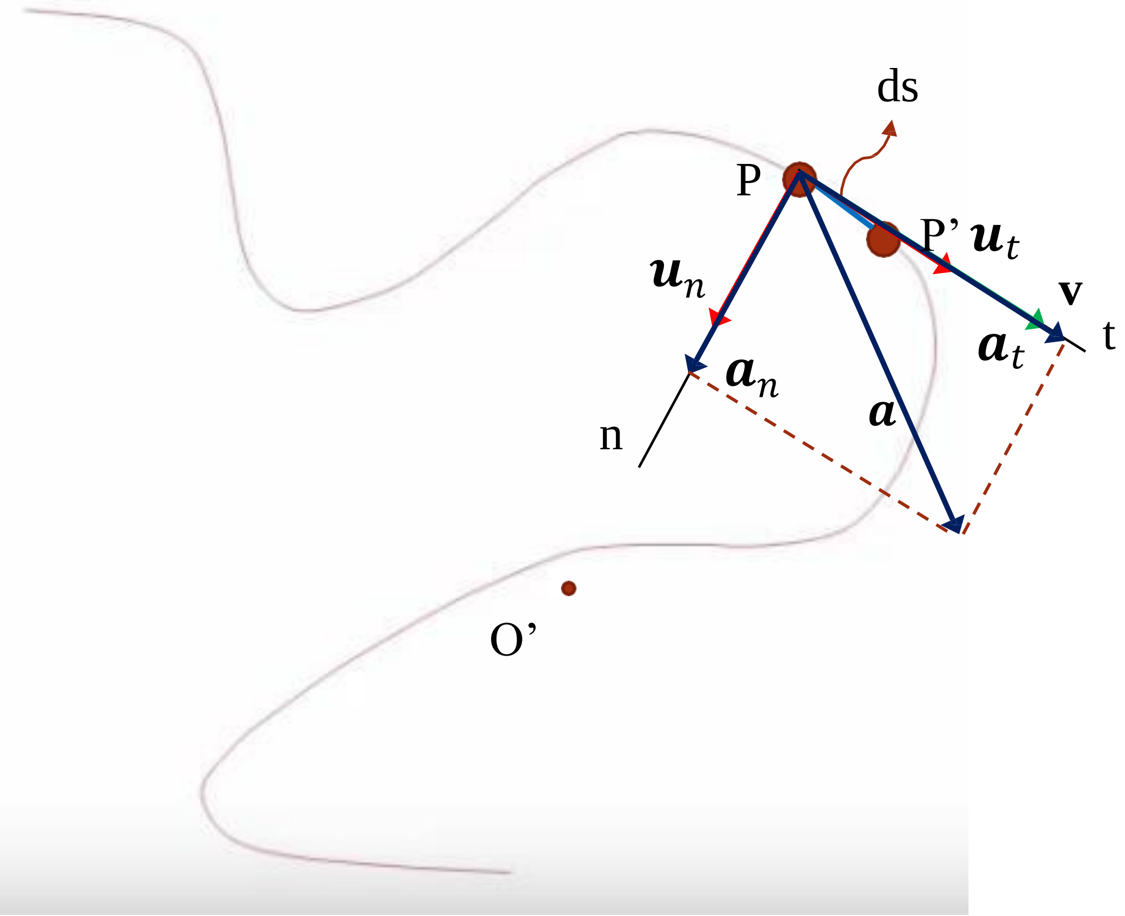
(a) the acceleration of each block,
(b) the tension in the cable.



Curvilinear Motion: Tangential & Normal Components



Particle Kinematics: Tangential & Normal Components



$$\boldsymbol{v} = v\boldsymbol{u}_t$$

$$\boldsymbol{a} = a_t\boldsymbol{u}_t + a_n\boldsymbol{u}_n$$

Here,

$$a_t = \frac{dv}{dt}$$

$$a_n = \frac{v^2}{\rho}$$

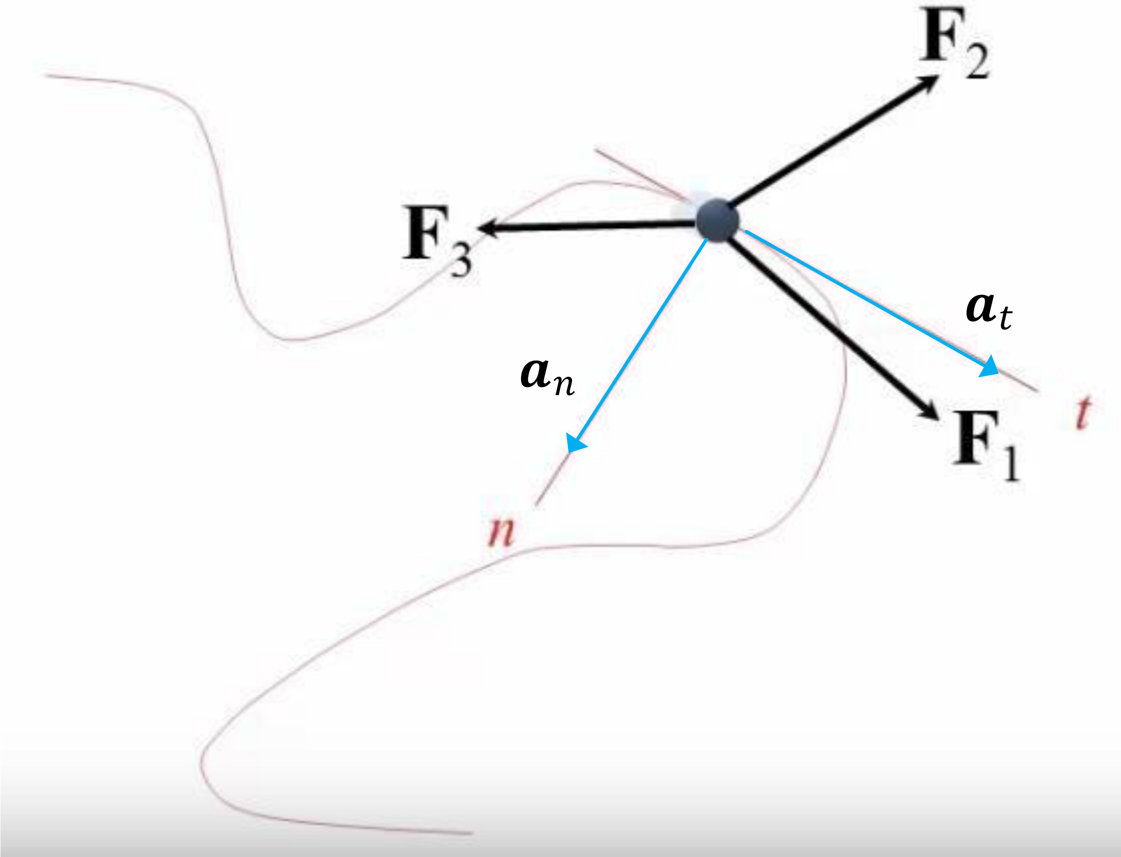
$$a = \sqrt{a_t^2 + a_n^2}$$

$\boldsymbol{Eq^n}$ of Motion: Tangential & Normal Components

$$\sum \boldsymbol{F} = m\boldsymbol{a}$$

$$\boldsymbol{\Sigma F_t = ma_t} \qquad \longrightarrow \qquad \boldsymbol{\Sigma F_t = m\frac{dv}{dt}}$$

$$\boldsymbol{\Sigma F_n = ma_n} \qquad \qquad \boldsymbol{\Sigma F_n = m\frac{v^2}{r}}$$

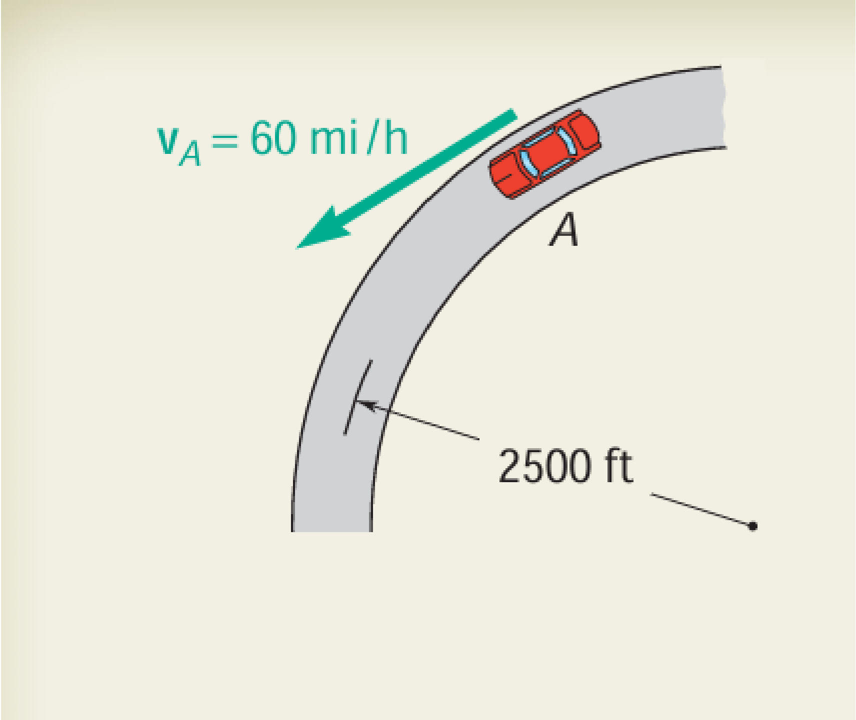


Kinematics

Problem:

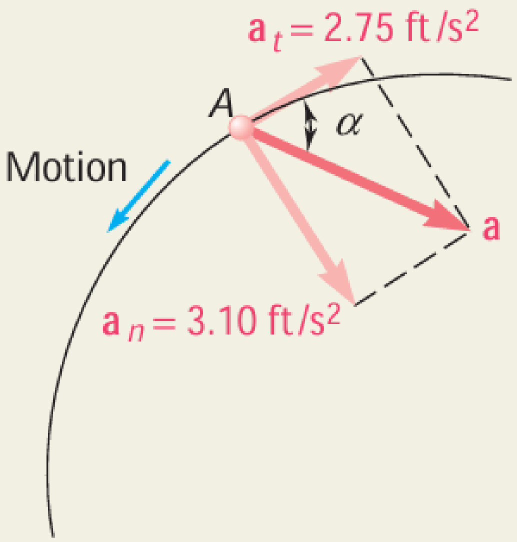
A motorist is traveling on a curved section of highway of radius 2500 ft at the speed of 60 mi/h. The motorist suddenly applies the brakes, causing the automobile to slow down at a constant rate. Knowing that after 8 s the speed has been reduced to 45 mi/h,

determine the acceleration of the automobile immediately after the brakes have been applied.



Reference: Example 11.10

Kinematics



SOLUTION

Tangential Component of Acceleration. First the speeds are expressed in ft/s.

$$60 \text{ mi/h} = \left(60 \frac{\text{mi}}{\text{h}}\right)\left(\frac{5280 \text{ ft}}{1 \text{ mi}}\right)\left(\frac{1 \text{ h}}{3600 \text{ s}}\right) = 88 \text{ ft/s}$$

$$45 \text{ mi/h} = 66 \text{ ft/s}$$

Since the automobile slows down at a constant rate, we have

$$a_t = \text{average } a_t = \frac{\Delta v}{\Delta t} = \frac{66 \text{ ft/s} - 88 \text{ ft/s}}{8 \text{ s}} = -2.75 \text{ ft/s}^2$$

Normal Component of Acceleration. Immediately after the brakes have been applied, the speed is still 88 ft/s, and we have

$$a_n = \frac{v^2}{\text{r}} = \frac{(88 \text{ ft/s})^2}{2500 \text{ ft}} = 3.10 \text{ ft/s}^2$$

Magnitude and Direction of Acceleration. The magnitude and direction of the resultant $\mathbf{a}$ of the components $\mathbf{a}_n$ and $\mathbf{a}_t$ are

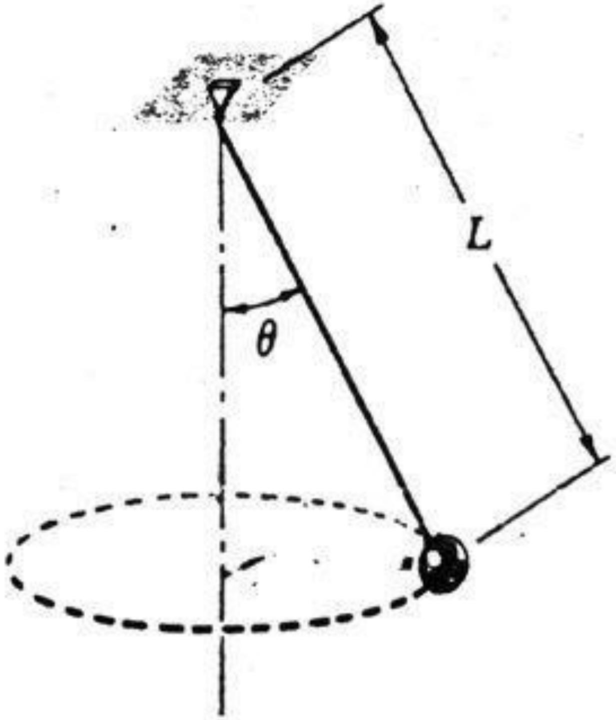
$$\tan \text{a} = \frac{a_n}{a_t} = \frac{3.10 \text{ ft/s}^2}{2.75 \text{ ft/s}^2} \qquad \text{a} = 48.4° \quad ◀$$

$$a = \frac{a_n}{\sin \text{a}} = \frac{3.10 \text{ ft/s}^2}{\sin 48.4°} \qquad \mathbf{a} = 4.14 \text{ ft/s}^2 \quad ◀$$

Equation of Motion

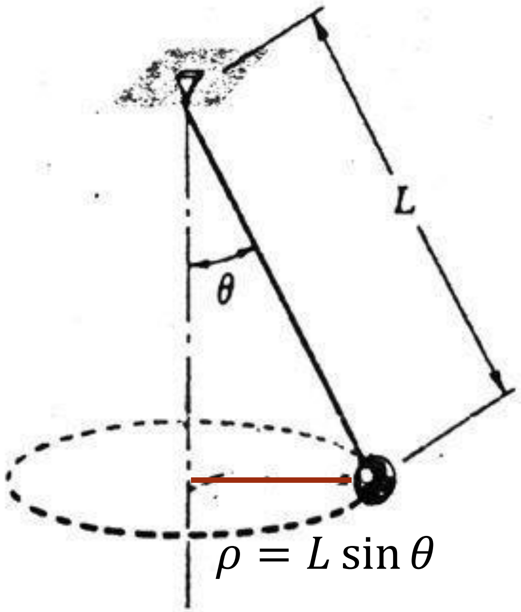
Problem:

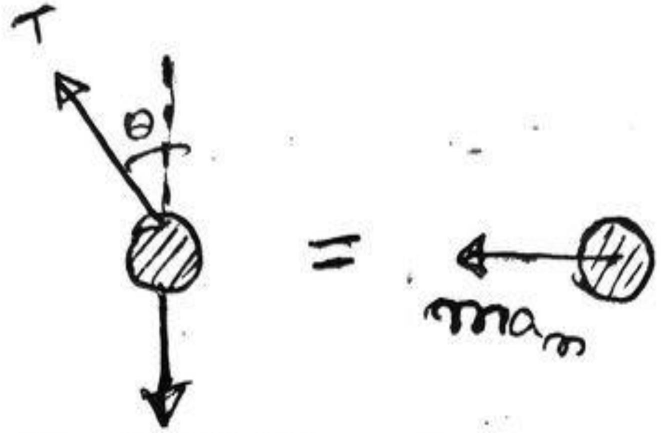
A 2-kg ball revolves in a horizontal circle as shown at a constant speed of $1.5\frac{m}{s}$. Knowing that $L = 600\,mm$, determine (a) the angle θ that the cord forms with the vertical, (b) the tension in the cord.



Equation of Motion

Solution:





$$W = 2 \times 9.81$$
$$= 19.62\,N$$

$$\overset{+}{\rightarrow} \Sigma F = ma$$
$$\Rightarrow -T\sin\theta = -2a_n$$
$$\Rightarrow T = \frac{2a_n}{\sin\theta}$$
$$\Rightarrow T = \frac{2v^2}{\rho\sin\theta} = \frac{2v^2}{L\sin^2\theta} \quad \text{(I)}$$

$$+\uparrow \Sigma F = ma$$
$$\Rightarrow T\cos\theta - W = 0$$
$$\Rightarrow T = \frac{19.62}{\cos\theta} \quad \text{(II)}$$

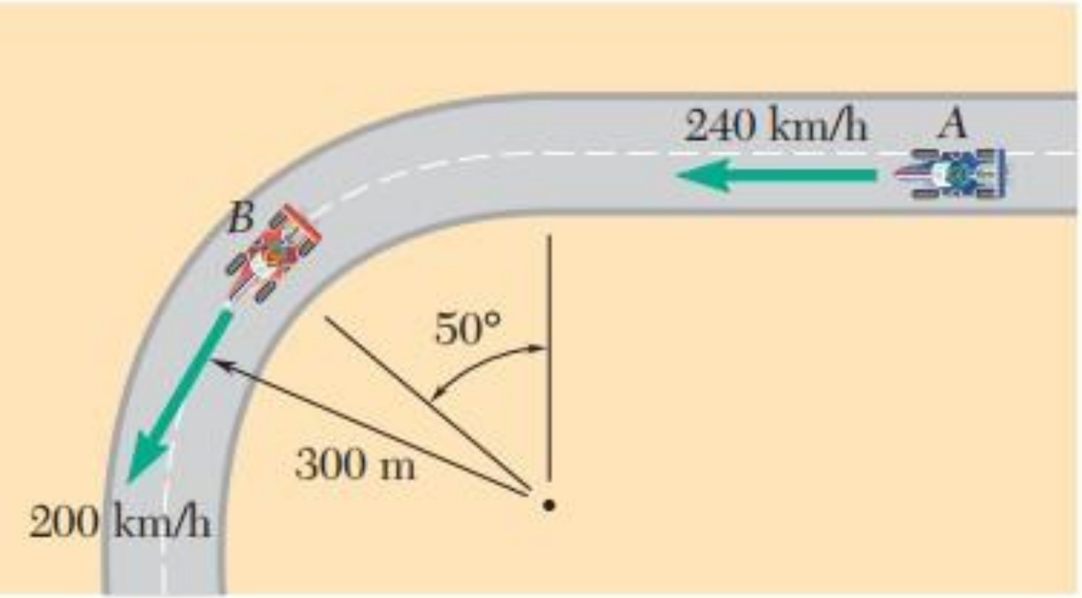
From (I) and (II) $\Rightarrow$

$$\theta = 34.21^\circ$$
$$\therefore T = 23.73\,N$$

Particle Kinematics: Tangential & Normal Components

Practice Problem:

Race car *A* is traveling on a **straight portion** of the track while race **car *B*** is traveling on a **circular portion** of the track. At the instant shown, the speed of ***A*** is **increasing** at the rate of $\mathbf{10\ m/s^2}$, and the speed of ***B*** is **decreasing** at the rate of $\mathbf{6 m/s^2}$. For the position shown, determine



$$\overline{v_A} = 240\frac{km}{h} = 66.67\frac{m}{s}$$

$$\overline{v_B} = 200\frac{km}{h} = 55.56\ m/s \quad 50$$

$$(a_A)_t = 10\frac{m}{s^2}$$

$$(a_B)_t = 6\frac{m}{s^2} \quad 50°$$

$$(a)\overline{v_{B/A}} = ?\ (b)\overline{a_{B/A}} = ?$$

(*a*) the velocity of *B* relative to *A*,

(*b*) the acceleration of *B* relative to *A*.